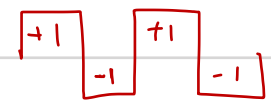
CDMA : each user is assigned a code with chips

orthogonal Codes S & T

Code 

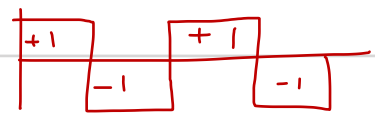
$\underbrace{\qquad\qquad\qquad}_{m \text{ chips}}$

* $S \cdot T = \frac{1}{m}\sum_{i=1}^{m} S_i T_i = 0$

$S \cdot \bar{T}$ is also

* $S \cdot S = \frac{1}{m}\sum_{i=1}^{m} S_i S_i = \frac{1}{m}\sum_{i=1}^{m} S_i^2 = 1$ and $S \cdot \bar{S} = -1$

Example

Station A chip sequence A 

B chip sequence B +1 +1 −1 −1

C chip sequence C +1 −1 −1 +1

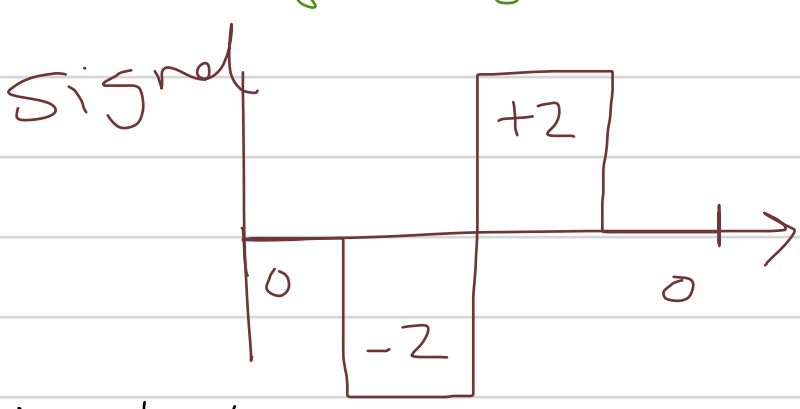
A is sending 1 bit (1) $\xrightarrow{\text{Code}}$ send +1 −1 +1 −1

B sending 1 bit (0) $\xrightarrow{\text{code complement}}$ send −1 −1 +1 +1

C sending nothing $\longrightarrow$ nothing

$\oplus$

Signal

Signal 

Detection:

Data for A $= [0\ -2\ +2\ 0] \cdot [+1\ -1\ +1\ -1] = \frac{}{4}$

$= [0 + (+2) + (+2) + 0]/4 = 1$ set binary 1

Data for B $= [0\ -2\ +2\ 0] \cdot [+1\ +1\ -1\ -1]/4 = \frac{-4}{4} = -1$ set binary 0

Data for C $= [0\ -2\ +2\ 0][+1\ -1\ -1\ +1]/4 = 0 \rightarrow$ Did not send any.